

LIBRARY MANAGEMENT: -

Q1. Write an SQL query to list all students who belong to the Computer Science department.

Q2. Write an SQL query to display the titles of all books currently not returned.

Q3. Write an SQL query to find the names of students who borrowed books in February 2024.

Q4. Write an SQL query to show the total number of books borrowed by each student.

Q5. Write an SQL query to retrieve the title, author, and student name for each borrowed book (include returned and not returned).

Q6. Write an SQL query to find the book(s) borrowed the most times.

Q7. Write an SQL query to list students who have never borrowed any book.

Q8. Write an SQL query to count the number of books in each category.

Answers: -

```
1 • create database lib;
2 • use lib;
3
4 • CREATE TABLE Students (
5     student_id INT PRIMARY KEY,
6     name VARCHAR(50),
7     department VARCHAR(50),
8     year INT
9 );
10
11 • CREATE TABLE Books (
12     book_id INT PRIMARY KEY,
13     title VARCHAR(100),
14     author VARCHAR(50),
15     category VARCHAR(30)
16 );
--
```

```

18 • CREATE TABLE BorrowedBooks (
19     borrow_id INT PRIMARY KEY,
20     student_id INT,
21     book_id INT,
22     borrow_date DATE,
23     return_date DATE,
24     FOREIGN KEY (student_id) REFERENCES Students(student_id),
25     FOREIGN KEY (book_id) REFERENCES Books(book_id)
26 );

```

```

28 • SELECT *
29 FROM Students
30 WHERE department = 'Computer Science';
31
32 • SELECT b.title

```

Result Grid | Filter Rows: | Edit: | Export/Import: |

	student_id	name	department	year
*	NULL	NULL	NULL	NULL

```

32 • SELECT b.title
33 FROM Books b
34 JOIN BorrowedBooks bb ON b.book_id = bb.book_id
35 WHERE bb.return_date IS NULL;

```

Result Grid | Filter Rows: | Export: | Wrap Cell

title

```

37 • SELECT DISTINCT s.name
38 FROM Students s
39 JOIN BorrowedBooks bb ON s.student_id = bb.student_id
40 WHERE bb.borrow_date BETWEEN '2024-02-01' AND '2024-02-29';
41

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

name

```

42 • SELECT s.student_id, s.name, COUNT(bb.book_id) AS total_borrowed
43 FROM Students s
44 LEFT JOIN BorrowedBooks bb ON s.student_id = bb.student_id
45 GROUP BY s.student_id, s.name;

```

```

47 • SELECT b.title, b.author, s.name

```

Result Grid		
student_id	name	total_borrowed

```

47 • SELECT b.title, b.author, s.name
48 FROM BorrowedBooks bb
49 JOIN Books b ON bb.book_id = b.book_id
50 JOIN Students s ON bb.student_id = s.student_id;

```

Result Grid		
title	author	name

```

52 • SELECT b.title
53 FROM Books b
54 JOIN BorrowedBooks bb ON b.book_id = bb.book_id
55 GROUP BY b.book_id, b.title
56 HAVING COUNT(*) = (
57     SELECT MAX(borrow_count)
58     FROM (
59         SELECT COUNT(*) AS borrow_count
60         FROM BorrowedBooks
61         GROUP BY book_id
62     ) AS t
63 );

```

Result Grid	
title	

```

65 • SELECT s.student_id, s.name
66 FROM Students s
67 WHERE s.student_id NOT IN (
68     SELECT student_id FROM BorrowedBooks
69 );

```

```

71 • SELECT category, COUNT(*) AS total_books

```

Result Grid

student_id	name
NULL	NULL

```

71 • SELECT category, COUNT(*) AS total_books
72 FROM Books
73 GROUP BY category;
74

```

Result Grid

category	total_books
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